BA ZC581 :
Applied Data Science
Module: 7

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Disclaimer

- The slides presented here are obtained / extracted from the authors of the prescribed books and from various other subject contributors.
- I hereby thank and acknowledge all the contributors for their material and inputs.
- I have added and modified a few slides to suit the requirements of the course.

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Module 7: Data Transformation and Normalization

Module 7 Topics

Topic 1: Normalization

Topic 2: Binning

Topic 3: Log/Square root Transform

Topic 4: Z-Score

Topic 5: Min-Max Scaling

Ref: T1 , Ch 5,6

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Topic 1: Normalization

Normalization

- In data preprocessing, normalization is a feature scaling technique that resizes numerical values into a common range (often 0–1).
- It ensures that features with different scales or units contribute equally to machine learning models, improving accuracy and convergence speed.
  - Imagine employee table with age and income as two numeric columns, each having values in different ranges (18-75 for age and 2,00,000 – 40,00,000 for salary). Using these values in computations causes large gradients causing the model to malfunction.
  - Bring the values to a common scale, 0-1

Normalization

- In essence, it is transforming raw feature values to a standard scale without distorting their relative relations.
- Standard range: [0-1], [-1,+1]
- Why it matters?
  - Equal contribution of features: Ensures no single variable outweighs others due to scale differences.
  - Improves performance: Especially important for distance-based algorithms like k-Nearest Neighbors (KNN), K-Means clustering, and Support Vector Machines (SVM).
  - Accelerates convergence: Gradient-based models (e.g., neural networks, logistic regression) train faster when features are normalized.
  - Better interpretability: Makes feature comparisons and visualizations more meaningful

Common Techniques

Common techniques used are:

Type	How It Works	Range/Scale	Best Use Cases
Min-Max Normalization	Rescales values to a fixed range (commonly [0,1] or [-1, +1])	Fixed (e.g., 0–1)	Distance-based algorithms (KNN, K-Means), image pixel scaling
Z-Score Normalization	Centers data around mean 0 with standard deviation 1	Not fixed	Regression, SVM, neural networks; robust to different distributions
Decimal Scaling	Moves decimal point based on max absolute value	Depends on data	Simple datasets where scaling by powers of 10 suffices
Log Transformation	Applies logarithm to compress large ranges and reduce skewness	Depends on log base	Skewed data, exponential growth patterns, financial or scientific datasets
Unit Vector Normalization	Scales feature vector to have unit length (magnitude = 1)	Vector length = 1	Text mining, cosine similarity, NLP embeddings

Use Min-Max when visualizing high variance data.
Apply Z-Score when comparing anomaly detection signals across different systems.
Try Log Transformation for skewed transaction data to highlight meaningful variations.

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Topic 2: Binning

Binning

- Data binning, also called data discrete binning or data bucketing, is a data pre-processing technique used to reduce the effects of minor observation errors.
- Values within a range are grouped (ex: all people with income less than 3lpa are grouped into low-income bin)
- The original data values which fall into a given small interval, a bin, are replaced by a value representative of that interval, often a central value
- Binning may regularize the data and may lead to loss of information. (Quantization noise)

Simple Example

Suppose we have a dataset of shoulder sizes (in inches):

[15,16,17,18,19,20,21,22,23,24]

Define T-shirt bins as follows:

Size	Shoulder Size Range (inches)
S	15 – 16.9
M	17 – 18.9
L	19 – 20.9
XL	21 – 24

Applying binning:

15 → **S**

16 → **S**

17 → **M**

18 → **M**

19 → **L**

20 → **L**

21 → **XL**

22 → **XL**

23 → **XL**

24 → **XL**

Binning converts **continuous numeric data (shoulder size)** into **categorical labels (shirt sizes)**.

Binning

- Binning Types:
  - Numerical
  - Categorical

Numerical Binning

- Used to bin numeric features in the data; can also be used as a classification technique (the T-shirt example earlier)

Type	Description	Pros	Cons
Equal-Width Binning	Divides the range into bins of equal size.	Simple to implement	May produce uneven bin populations
Equal-Frequency Binning (also called Equal Depth)	Each bin contains roughly the same number of data points.	Balanced bin sizes	Bin widths may vary widely
Custom Binning / Rule based binning	Bins are manually defined based on domain knowledge or statistical rules.	Highly flexible and interpretable	Requires expertise; risk of bias

- Elements in the bin can define the bin property

Numeric Binning

- Equal-width (distance) partitioning:
  - It divides the sample range into ***N*** intervals of equal size: uniform grid
  - if *A* and *B* are the lowest and highest values of the attribute, the width of intervals will be:
  - $$W = (B-A) / N$$
  - The most straightforward computation
  - But *outliers may dominate presentation*
  - *Skewed data* is not handled well.

Equal width partitioning

- Data for a unit price (in dollars):
  4, 8, 29, 15, 24, 21, 21, 28, 26, 25, 9, 34
- Range : 34-4 = 30
- No of partitions: 4
- Width of partition, W = 30/4 = 7
- Partitions :
  - B1: 4 - 11 : 4,8,9
  - B2: 12 – 19 : 15
  - B3: 20 – 27 : 21, 21, 24,25,26
  - B4: 28 – 34 : 28, 29, 34

Equal Depth partitioning

- Equal-depth (frequency) partitioning:
  - It divides the sample range into *N* intervals each which contain equal number of samples:
  - If the data has M samples, and we need N bins, the size of the bins = M/N
  - Width of the bins (W) is not fixed.

Equal Depth Partitioning

- Data for a unit price (in dollars):
  4, 8, 29, 15, 24, 21, 21, 28, 26, 25, 9, 34
- Sorted data for a unit price (in dollars):
  4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34
- M=12
- No of Partitions: N= 4.
- Size of each partition k = 3 ; pick k items from sorted list and add to the partition.
- Partitions :
  - B1: 4, 8, 9,
  - B2: 15,21, 21
  - B3: 24,25, 26,
  - B4: 28, 29, 34

Custom Binning / Rule Based binning

- Used also for assigning class labels to data
- Data for a unit price (in dollars):
  4, 8, 29, 15, 24, 21, 21, 28, 26, 25, 9, 34
- Binning rules:

Values between 0-5 : Bin1 – very low values
Values between 6-10: Bin2 – low values
Values between 11-20: Bin3 – medium values
Values between 21-30: Bin4 – high values
Values greater than 30: Bin 5 – very high values

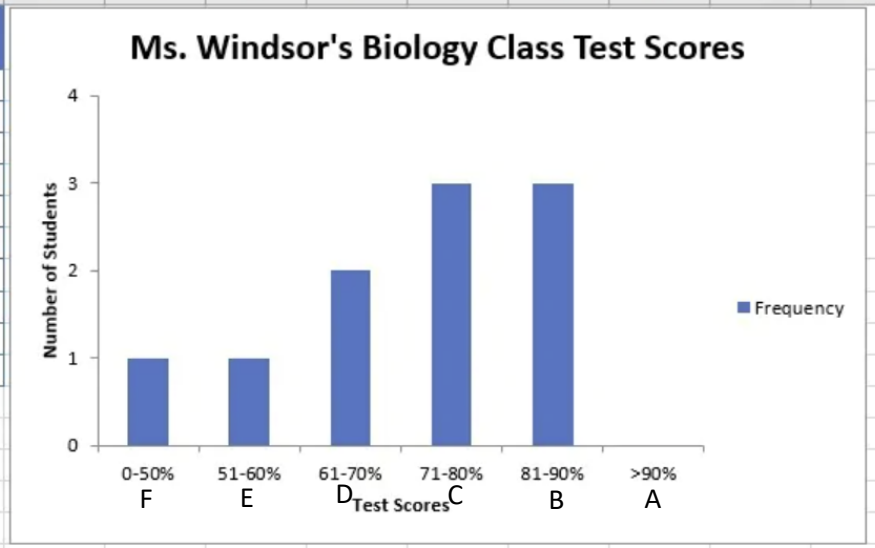
- Partitions:
  - B1: 4
  - B2: 8,9
  - B3: 15
  - B4: 29,24,21,28,26,25
  - B5: 34
- 4, 8, 29, 15, 24, 21, 21, 28, 26, 25, 9, 34
- VL,L, H, M, H, H, H, H, H, H, L, VH

Custom / Rule based binning

RULEBASE:

BinF: [0,50]
BinE: [51,60]
BinD: [61,70]
BinC: [71,80]
BinB: [81,90]
BinA: [91,100]

Biology Scores
41
56
63
68
72
78
79
81
85
85



E

Binning for Data Smoothening

Sorted data for a unit price (in dollars):

4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

* Partition into (equi-depth) bins: Assume 3 bins.
  - Bin 1: 4, 8, 9, 15 bin boundaries: [4,15]
  - Bin 2: 21, 21, 24, 25 bin boundaries: [21,25]
  - Bin 3: 26, 28, 29, 34 bin boundaries: [26,34]
* Smoothing by bin means:
  - Bin 1: 9, 9, 9, 9 mean=(4 + 8 + 9 + 15) /4=9
  - Bin 2: 23, 23, 23, 23 mean=(21+ 21+ 24 + 25)/4 =23
  - Bin 3: 29, 29, 29, 29 mean=(26 + 28 + 29 + 34)/4=29
* Smoothing by bin boundaries: [4,15],[21,25],[26,34]
  - Bin 1: 4, 4, 4, 15
  - Bin 2: 21, 21, 25, 25
  - Bin 3: 26, 26, 26, 34

Move a value to the nearest bin boundary

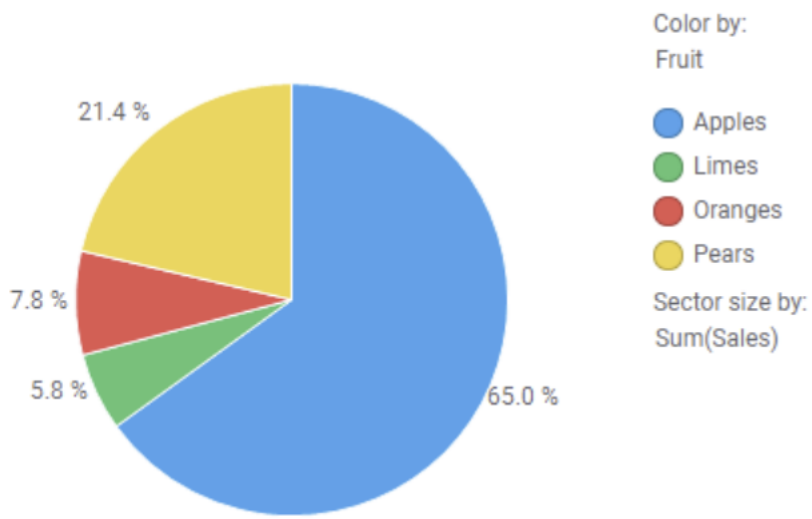
Binning Categorical Data

- Suppose you have a survey data on favorite fruits : ["Apple", "Banana", "Mango", "Orange", "Strawberry", "Blueberry", "Pineapple", "Watermelon"]
- Define bins (groups)
  - **Bin 1: Citrus Fruits** → ["Orange"]
  - **Bin 2: Berries** → ["Strawberry", "Blueberry"]
  - **Bin 3: Tropical Fruits** → ["Mango", "Pineapple", "Banana"]
  - **Bin 4: Others** → ["Apple", "Watermelon"]
- Replace values with Bin labels:

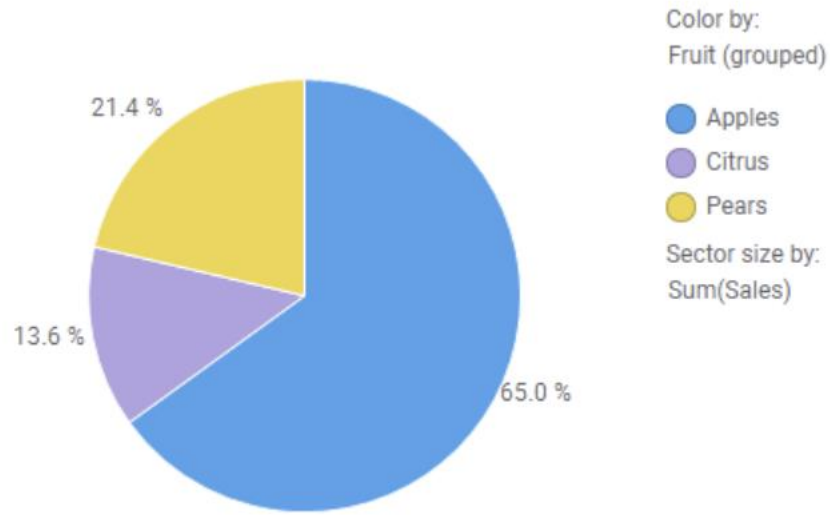
  ["Others", "Tropical Fruits", "Tropical Fruits", "Citrus Fruits", "Berries", "Berries", "Tropical Fruits", "Others"]
- Reduces complexity: Instead of 8 categories, you now have 4 bins. Used for dimensionality reduction
- Another example: **Cities** data values can be binned to **States**

Binning categorical Data

The pie chart shows sales per apples, limes, oranges and pears.



Below oranges and limes have been grouped into a bin called “Citrus”.



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Topic 3: Log / Square Root Transformation

Log / Square Root Transformation

- Both help reduce skewness, compress large ranges, and stabilize variance in datasets — especially useful for handling outliers or exponential growth.
- Distribution becomes normal after the transformations
- Transformed values lose meaning

Log Transformation

- Applying a logarithmic function to each data value
  e.g., $y_i = \log_b(x_i)$, where b is the base, often *e* or 10
- Example:
  - Original values = [10, 200, 1000, 3000]
  - After $\log_{10}$ transformation = [1, 2.3013, 3, 3.477].
- Natural log (ln): Common in statistics, uses base e (approx 2.718)
- Restriction: Only works on positive values. Zero or negative values must be adjusted (e.g., adding a constant).
- Typically, we perform log(x+1) transformation for value range >=0

Square root Transformation

- A power transformation where data values are replaced by their square root
- Apply square root to each value

  e.g., $y_i = \sqrt{X_i}$
- Example:
  - Original values = [1, 4, 9, 16, 100]
  - After transformation = [1, 2, 3, 4, 10].
- Compresses large values while spreading out smaller ones, but less aggressively than log transformation.
- Often used when log transformation is too strong or data contains many zeros. (more than 20%)

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Topic 4: Z-Score

Standard Deviation

- Determines volatility
- It indicates how far from the average the data spreads.
  - This helps you determine the limitations and risks Inherent in decisions based on that data.

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

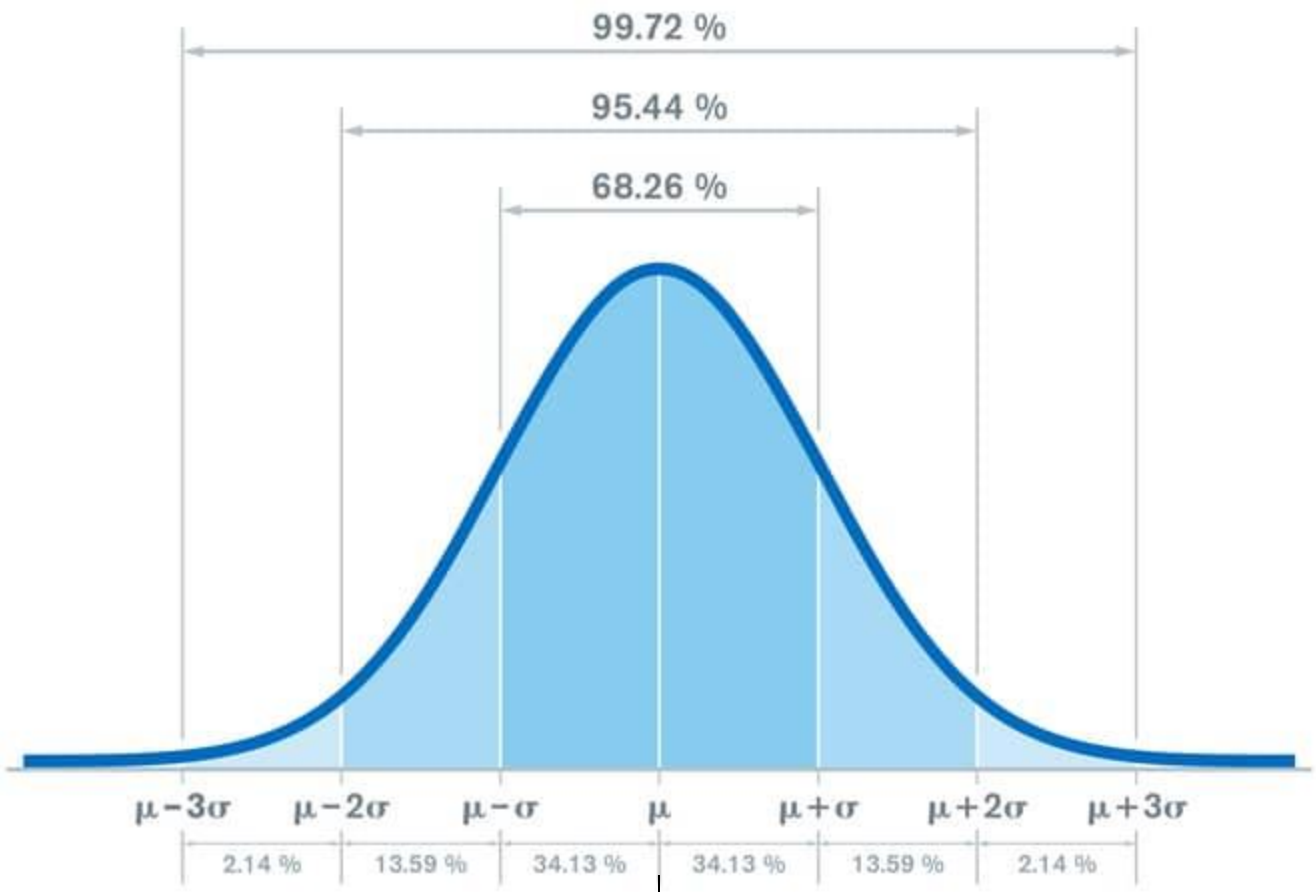
σ = population standard deviation

N = the size of the population

x_i = each value from the population

μ = the population mean

Normal Distribution Curve



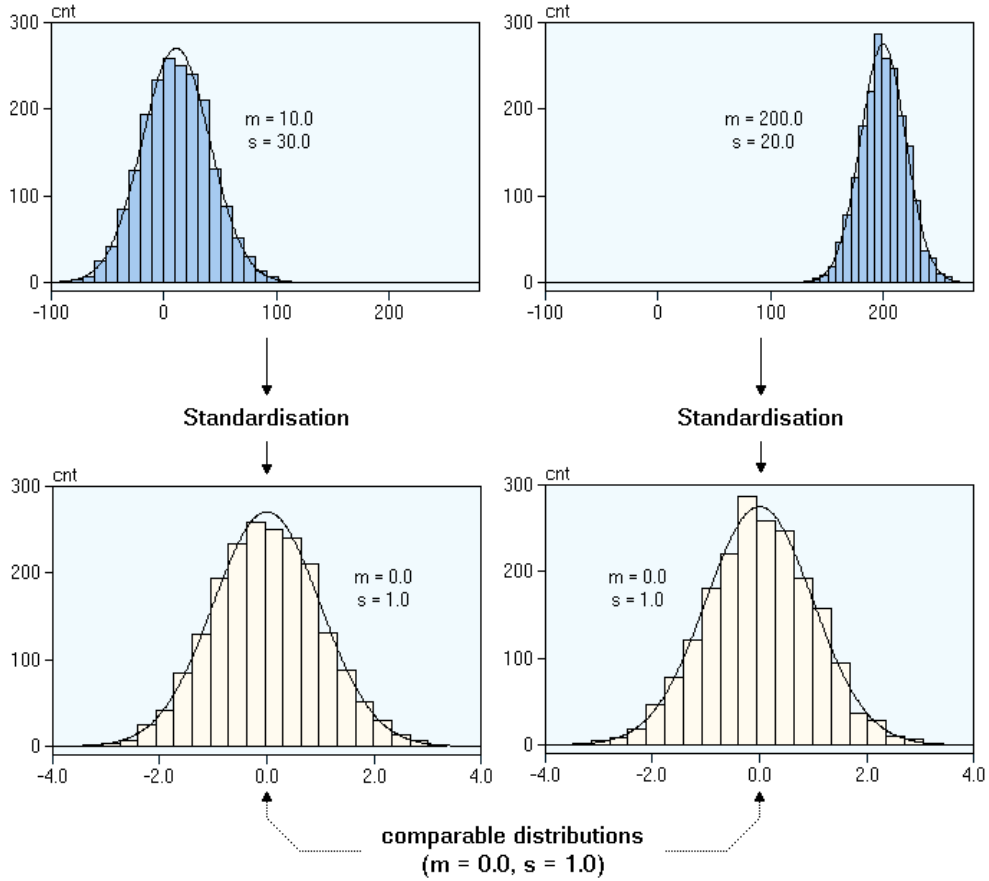
Mean
Median
Mode

Z-Score

- Z-Score (also called the Standard score) measures how many standard deviations a data point is away from the mean.
- **The z-score allows us to determine how usual or unusual a data point is in a distribution**.
- The z-score is calculated for each value in a given feature. It takes into account the standard deviation and the mean of the feature.
- **z** = $(X - \mu) / \sigma$
  - where:
  - X is a single raw data value
  - μ is the population mean
  - σ is the population standard deviation
- For example, If the mean score is 65 and the SD is 10, for a student scoring 60, the z-score = (60 -65)/10 = -0.5
  - This is 17% below the mean; he is in the 33 percentile.
- A large positive value of z-score (>3.5) or a large negative value of z-score (< -3.5) indicates an outlier.

Z-Score Standardization

variables to have a zero mean and standard deviation of one.



Z-Score example

- My time to run the First 200m race is 28 seconds. The mean time for this race is 31 seconds and the standard deviation is 1.5 seconds.
- My time to run the Second 500m race is 132 seconds. The mean time for this race is 125 seconds and the standard deviation is 8.2 seconds.
- Let's calculate the z-score in each:
  - For the 200m race, the z-score is -2
  - For the 500m race, the z-score is 0.854
- In the First 200m run, the timing was better than the average runner and hence performance is better than the Second race.

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Topic 5: Min-Max Scaling

Min-Max Scaling

- Min-Max scaling is a normalization technique in data preprocessing that rescales features to a fixed range, usually [0,1].
- It works by subtracting the minimum value of the feature and dividing by the range (max–min), ensuring all values fall proportionally within the chosen interval.
  - The min value is assigned 0 and the maximum is assigned 1 and other values are placed linearly.
- However, it is sensitive to outliers, since extreme values can compress the majority of data into a narrow band

A clear example

- Ex1: Consider the following two features:
  - Feature A's lowest value is -0.5 and highest is +0.5.
  - Feature B's lowest value is -5.0 and highest is +5.0.
  - Feature A and Feature B have relatively narrow spans. However, Feature B's span is 10 times wider than Feature A's span.
- Hence, at the start of training, the model assumes that Feature B is ten times more "important" than Feature A.
  - Training will take longer than it should.
  - The resulting model may be suboptimal.
  - The overall damage due to not normalizing will be relatively small; however, it is good to normalize Feature A and Feature B to the same scale, perhaps -1.0 to +1.0.
- Ex2: Now consider two features with a greater disparity of ranges:
  - Feature C's lowest value is -1 and highest is +1.
  - Feature D's lowest value is +5000 and highest is +1,000,000,000.
- If we don't normalize Feature C and Feature D, the model will likely be suboptimal and training will take much longer to converge or even fail to converge entirely!

Linear Scaling calculations

Converting numeric values into a linear scale of 0-1

$$x' = (x - x_{min})/(x_{max} - x_{min})$$

where:

- x' is the scaled value.
- x is the original value.
- x_{min} is the lowest value in the dataset of this feature.
- x_{max} is the highest value in the dataset of this feature.

Also called min-max scaling

For example, consider a feature named **quantity** whose natural range spans 100 to 900.
Suppose the natural value of **quantity** in a particular example is 300.
We can calculate the normalized value of 300 as follows:

- x = 300
- x_{min} = 100
- x_{max} = 900

```
x' = (300 - 100) / (900 - 100)
x' = 200 / 800
x' = 0.25
```

Min-Max Scaling : Another example

This method linearly transforms data into a fixed range, usually $[0, 1]$.

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

- x = original value
- $x_{\min}$ = minimum value in the dataset
- $x_{\max}$ = maximum value in the dataset
- x' = normalized value (between 0 and 1)

👉 Example: If exam scores range from 30 to 95, and a student scored 60:

$$x' = \frac{60 - 30}{95 - 30} = \frac{30}{65} \approx 0.46$$

Linear Scaling suitability

- Suppose **human_age** is a feature. Linear scaling is a good normalization technique for age because:
  - The approximate lower and upper bounds are 0 to 100.
  - age contains a relatively small percentage of outliers. Only about 0.3% of the population is over 100.
  - Although certain ages are somewhat better represented than others, a large dataset should contain sufficient examples of all ages.

- Suppose our model has a feature named **net_worth** that holds the net worth of different people. Would linear scaling be a good normalization technique for **net_worth**?
  - Linear scaling would be a poor choice for normalizing **net_worth**. This feature contains many outliers, and the values are not uniformly distributed across its primary range. Most people would be squeezed within a very narrow band of the overall range.

Min-Max with & without Outlier

Without Outlier

- Consider sample data: [5,10,15,20,25,30,35,40,45,50]
- Minimum = 5 ; Maximum = 50
- Formula: (x-5)/(50-5)

Original Value	Scaled Value
5	0.000
10	0.111
15	0.222
20	0.333
25	0.444
30	0.556
35	0.667
40	0.778
45	0.889
50	1.000

Original values are evenly spread

With Outlier

- Consider sample data: [5,10,15,20,25,30,35,40,45,50, 1000]
- Minimum = 5 ; Maximum = 1000
- Formula: (x-5)/(1000-5)

Extended Value	Scaled Value
5	0.000
10	0.005
15	0.010
20	0.015
25	0.020
30	0.025
35	0.030
40	0.035
45	0.040
50	0.045
1000	1.000

Original values are compressed near 0

Others..

Decimal Scaling

Moves the decimal point based on maximum absolute value.

Moves the decimal point based on the maximum absolute value.

$$x' = \frac{x}{10^j}$$

- j = smallest integer such that $|x'| < 1$ for all values

👉 Example: If the largest value in the dataset is 987, then $j = 3$:

$$x' = \frac{987}{10^3} = 0.987$$

Unit Vector Normalization

Scales a vector so its length (magnitude) becomes 1.

$$x' = \frac{x}{\|x\|}$$

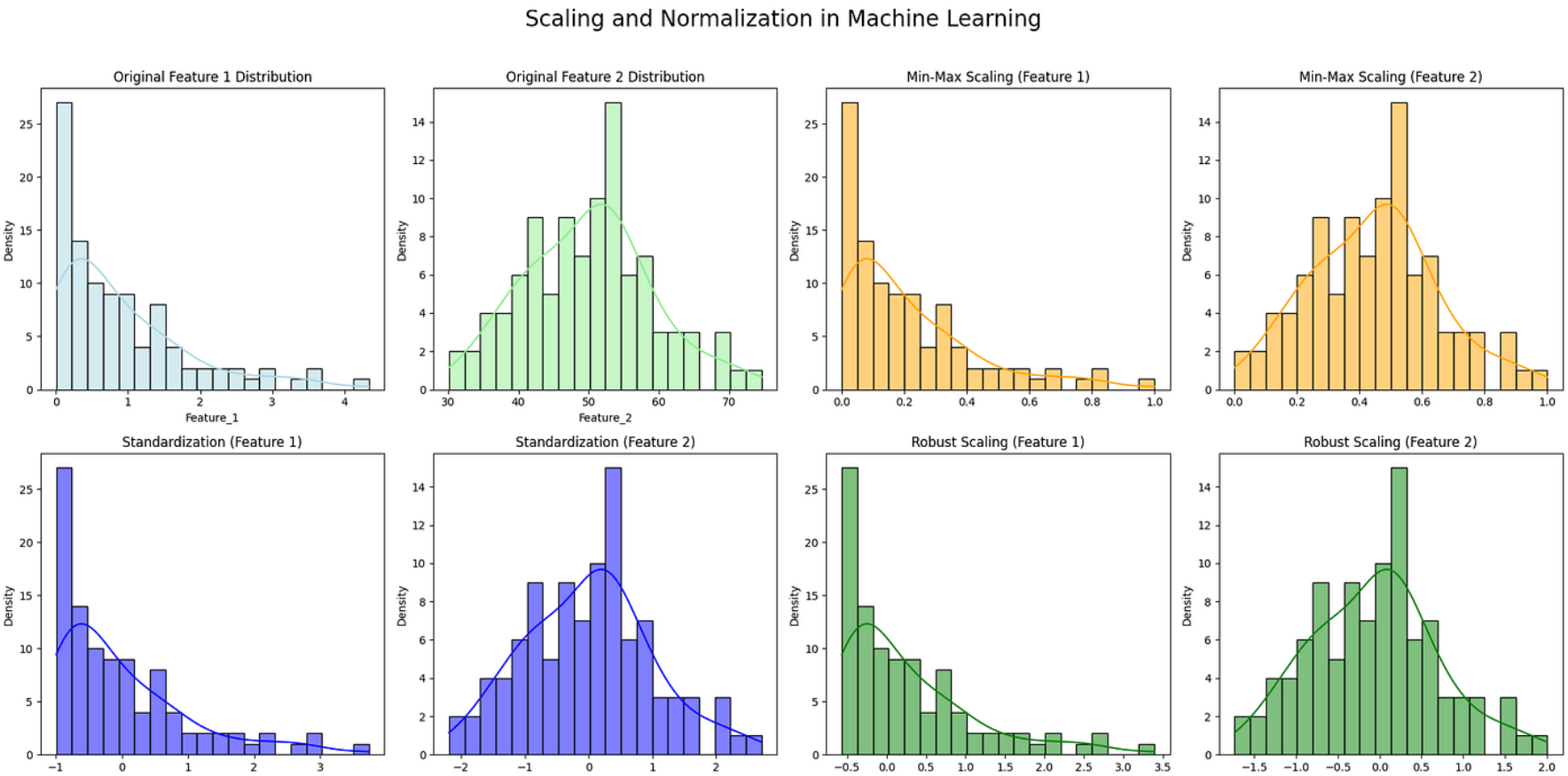
- $\|x\|$ = Euclidean norm (length of the vector)
- Useful in text mining and cosine similarity.

👉 Example: For vector $x = [3, 4]$:

$$\|x\| = \sqrt{3^2 + 4^2} = 5$$

$$x' = \left[\frac{3}{5}, \frac{4}{5}\right] = [0.6, 0.8]$$

Comparison of a few techniques



Robust Scaling: Centers data around median, scales using IQR

medium.com

Comparison – numerical data

Assume data : [30,45,60,75,90]

Original Score	Min-Max	Z-Score	Decimal Scaling	Log (Base 10)	Unit Vector
30	0.00	-1.26	0.30	1.48	0.19
45	0.25	-0.63	0.45	1.65	0.29
60	0.50	0.00	0.60	1.78	0.39
75	0.75	0.63	0.75	1.88	0.49
90	1.00	1.26	0.90	1.95	0.58

Summary

- Discussed Various Types of Transformation techniques used in Data Preprocessing
- Understood binning operations and their types
- Studied the need for log / square root transformations
- Understood Z-Score and its significance
- Understood min-Max scaling and how to use Min-Max scaling
- Studied a few implementation demos

Questions?